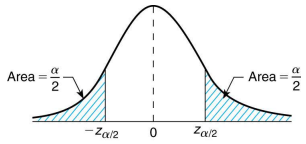
Agenda

- Earlier
  - Confidence intervals for proportions
  - Comparing means for two populations or treatments
- Today
  - Confidence intervals for proportions from two different populations

$$\Pr\left(-z_\alpha \leq \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \leq z_\alpha\right)$$



1

Review

2

Example: Comparing two populations

- We measuring sulfhydryl concentration (mmol) in blood samples of those without and with rheumatoid arthritis (RA)
- Statistical question:
  - What is the difference in the concentration for those without and with RA?
  - We look for the confidence interval in the difference in means using the provided data

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

Course details from IITB for CS215

- The role of statisitcs.
- Graphical and numerical methods for describing and summarising data.
- Probability.
- Population distributions.
- Sampling variability and sampling distributions.
- Estimation using a single sample.
- Hypothesis testing a single sample.
- *Comparing two populations or treatments.*
- Simple linear regression and correlation.
- Case studies.

3

Review: Confidence interval for difference

TABLE 7.2 *100(1 − σ) Percent Confidence Intervals for* $\mu_1 - \mu_2$

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$
$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$
$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2,\, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Pooled estimator

If we do not know the variances, and whether the variances are equal, then, as an empirical measure, we use a conservative estimate with df=min(n-1, m-1)

4

Review: Proportions

CI is not a protection against fabricated or sloppy data. We assume random sampling with independent observations, and an unbiased instrument to measure the data

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm$ z standard errors applied to *proportions*
- This works well if the distribution is normal, and
  - "quite well" when the distribution is unknown provided n is "sufficiently" large
  - Example: np(1-p)>10

$$\Pr\left(-z_{0.025} \leq \frac{\hat{p}-p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq z_{0.025}\right) \approx 0.95$$

5

Example: Comparing two populations

- 68% (n=870) non-smokers get "good" grades compared to 41% (n=130) smokers.
- What can we conclude?
- Contrast a single population where we ask
  - Does the fraction 0.68 (for non-smokers) represent the true proportion in the population

Get good grades

percent

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

6

Review: Single population

$$\Pr\left(-z_\alpha \le \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \le z_\alpha\right)$$

- Does the fraction 0.68 (for non-smokers) represent the true proportion in the population?
  1. Is the normal approximation valid? 870*0.68*(1-0.68) > 10, so yes. Empirical rule
  2. $\Pr\left(-z_{0.025} \le \frac{\hat{p}-p}{se(\hat{p})} \le z_{0.025}\right) = 0.95$ and $z_{0.025} = 1.96$,
  3. $se(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.01582$
  4. Therefore $CI = \hat{p} \pm 1.96\,\widehat{se}(\hat{p})$
     $= 0.68 \pm 1.96(0.01582)$
     $= 0.68 \pm 0.03100$
  5. It is not unreasonable to expect 65% of non-smokers getting good grades in the population

Non-causality: Non-smoking is not a shortcut to getting good grades

7

Review: How big should my sample be?

- The margin is $z \times \sqrt{(}p(1-p)/n$
- If m is the margin of error, then
  - want $z \times \sqrt{(}p(1-p)/n \le m$
  - Implies if we guess a proportion p^* want $n \approx \frac{z}{m}^2 p^*(1-p^*)$
- Examples
  - Margin is 10%, p=0.3, n=81
  - Margin is 5%, p=0.3, n=323

8

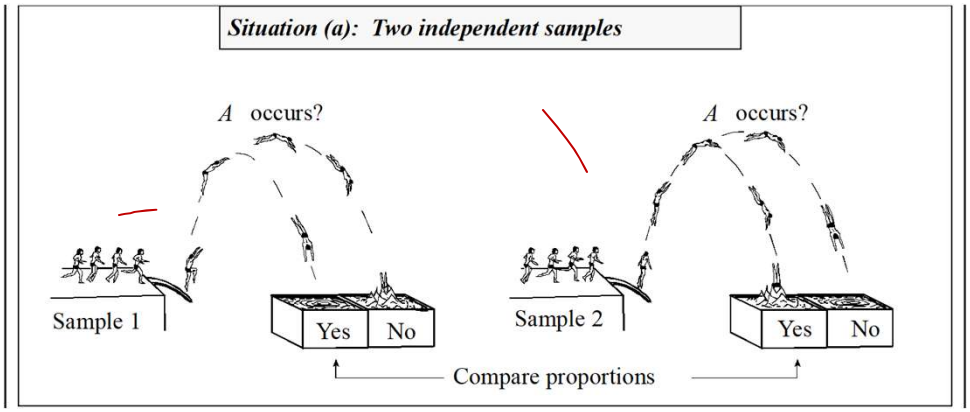
Example: Comparing two populations

- Good grades: 68% (n=870) non-smokers, 41% (n=130) smokers
- What can we conclude about the difference between the two populations?
  - Is the normal approximation tenable?
    - 130*0.41*(1-0.41) > 10? 870*0.68*(1-0.68) > 10, so yes.
  - Point estimate of the difference = 0.68-0.41, standard error = 0.045944
  - CI = $0.27 \pm 1.96 * 0.045944$= [0.18, 0.36]
  - Not unreasonable that non-smokers outperform smokers, the difference is 18 percentage points (with 95% confidence)

$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

9

Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Variance adds up

10

Example #2

- 18% difference in the level of support for Clinton. What's the CI?
- Level of support for Clinton in NJ and NY: may be considered independent
- Level of support for Clinton vs Dole in New Jersey: same population is being asked to make a choice
  - What's the difference?

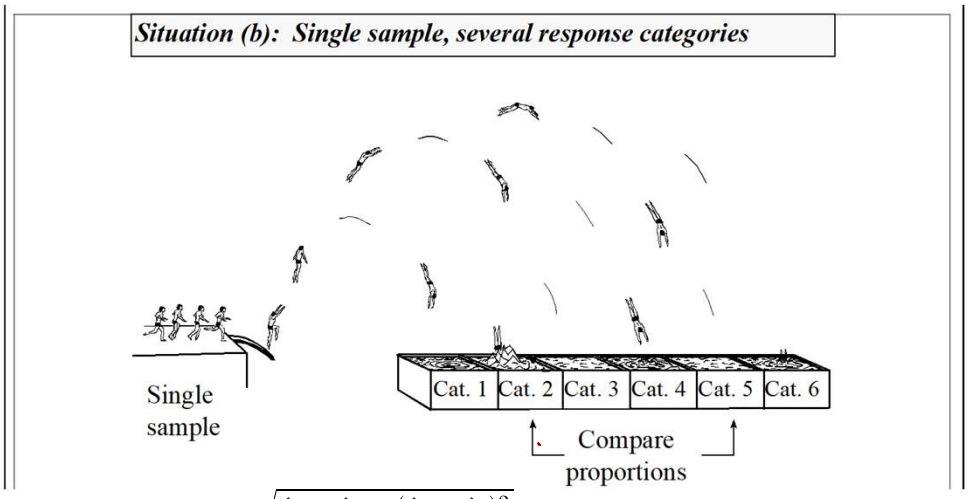
percent

State	n	Pre-election Polls				Election Results		
		Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996.

11

Proportions



$$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 + \hat{p}_2 - (\hat{p}_1 - \hat{p}_2)^2}{n}}$$

12

Example

- Level of support: for Clinton: 51 and for Dole: 33, difference is 18%.
- What is the confidence interval?
  - Answer: margin = 0.056, CI = [0.12, 0.24]

State	n	Pre-election Polls				Election Results		
		Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996 .

14

Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

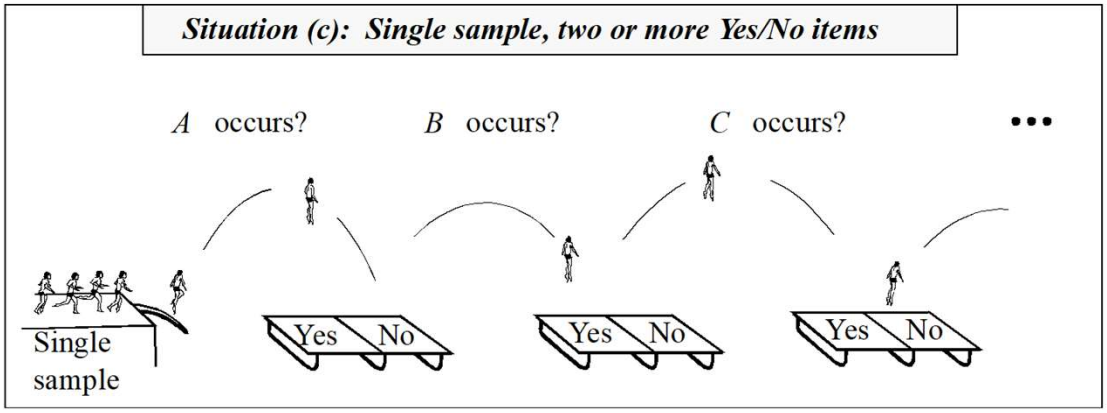
Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

15

Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\min(\hat{p}_1 + \hat{p}_2, \hat{q}_1 + \hat{q}_2) - (\hat{p}_1 - \hat{p}_2)^2}{n}} \qquad \hat{q}_1 = 1 - p_1, \hat{q}_2 = 1 - p_2$$

16

Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?
- Point estimate = 0.79 – 0.63
- n = 130 is large enough
- se = 0.065, [0.03, 0.29]

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)		Smoker	Nonsmoker
Get mostly A's or B's?		41	68
Read 1 or more hours/day?		54	72
Get drunk at least once a month?		63	10
Have smoked Marijuana?		79	14
Likely to try illegal drug in future?		42	14
	n	130	870

Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

17

Review

1. First verify if the normal approximation can be made
2. Check which case applies
   - Proportions from two independent samples
   - One sample of size n, several response categories
   - One sample of size n, many yes/no items
   - Calculate the standard error based on the formula
   - Use the normal approximation with the appropriate confidence level

18